Automated Modular Greenhouse

Here I will be describing the internal structure of the project.

As the name implies, this project is meant to be modular. Now the current version “0.1” can at most be considered a prototype, there are many flaws in the design, and its construction was not planned well enough. But that’s why we call it a prototype, and there is a lot to be learnt from it.

Hardware

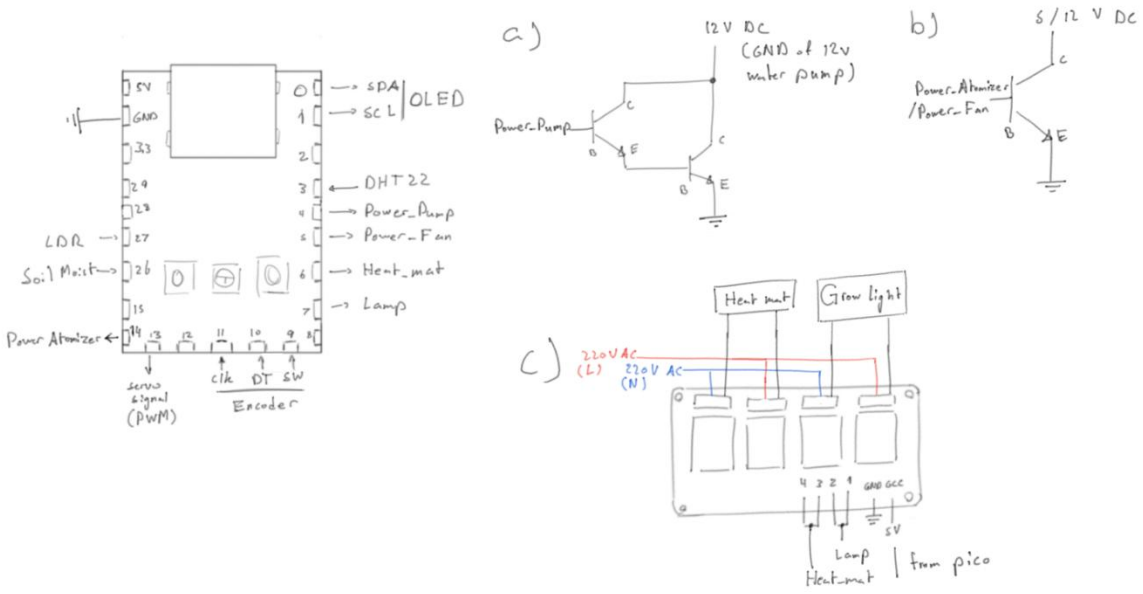
The control circuitry:

A RP2040-Zero is used as the brain for the project. It is connected to a Perfboard through Dupont terminals. The Perfboard has two terminals for power (5V and 12V) to drive peripherals.

In order to make the design as modular as possible none of the peripherals are directly soldered to the Perfboard, instead I used JST-XH connectors. Transistors were used to drive the higher power, intermittently active, peripherals (water pump, fan, ultrasonic atomizer) which would have otherwise burnt the microcontroller if driven directly by it. The rest (DHT22, OLED, LED, encoder, ldr, servo) are directly connected to their respective 5v pin from the JST connector. There are also 2 peripheral devices (grow light, heat mat) that run on mains 220V AC, for those I used a 4-relay module to control both the live and neutral of the port (which is an XT60) for added safety. Two relays take signal from a single wire so that all 4 relays are controlled via 2 GPIOs from the Pico.

Pictures of the underlying wiring on the board will not be shown as in all honesty it is a mess; however, I will provide a diagram to show the connections.

Inside the main unit are two power supplies, one for 5V and another for 12V, each rated at 2A. In hindsight, I should have used one and used a buck/ boost converter to drop or raise the voltage to save money and space.



Top left is the Pi-Pico zero (rp2040) and the components to which the pins connect to, an arrow pointing to the Pico is input from component, arrows pointing away from the pins means it an output to a component.

To power the Pico, I attached a USB-C cable to the 5v rails and connected it to the USB port on the Pico to power it. The reason behind this is that accessing the pad to power the Pico directly with 5V DC is too small and would have been too difficult to wire up.

Diagram “a” show the use of a Darlington transistor configuration in order to handle the current needed to drive the water pump, since a single transistor would have been at risk of burning out. However, it is possible to find transistors that can handle high current, removing the need to use two of them. Base of the left transistor is connected to “Power_Pump” through 1K ohm resistor. Both collectors are connected to negative terminals of Pump. Emitter of left transistor connected to base of right transistor whose emitter connects to gnd.

Diagram “b” is the circuit used for powering the atomizer and fan (two circuits, one for each component). For the atomizer collector connects to ground of the atomizer, the base is connected to the “Power_Atomizer” pin on Pico through a 1K ohm resistor, emitter is connected to the common ground of the circuit. A similar story with the fan power circuit, collector connects to ground of fan and Base connected to “Power_Fan” pin and emitter goes to common ground.

NOTE! Remember to add resistor between base and the pin on Pico, Max current pins can handle is in the mA (milli Amp) range, hence it is good practice to add resistor to avoid excessive current draw and damaging the MCU. 1 K ohm is good for most cases, but I suggest you do your own research.

Diagram “c” show how the mains voltage (220 V AC) components are connected. I used a 4-relay module to power the heat mat and grow light. It would have been possible to just use 2 relays but that would mean that I would be unable to fully disconnect all terminals connecting to the heat mat and grow light, making it less safe (which I wished to guarantee). The relays are setup in pairs, one for heat mat and another for grow light. Each pair has one relay connected to the live terminal (red) and the other to neutral (blue) terminal. These wires are connected to the common terminal of the relay (relays usually are set up in “Normally open”, “Common”, “Normally Closed”), and the wires that go to the mat and light connect to the “Normally Open” terminals. The wires (for the common terminal) are routed to the C13 socket that then connects with mains power.

I will state that using a C13 socket is over kill for powering the system, the system draws below 150 W (grow light draws the most at 100W) while C13 is rated for up to 2500W (at 250 V), so it is possible to use smaller connectors such as C7 (which is rated for up to 250 W at 250 V) to save on costs.

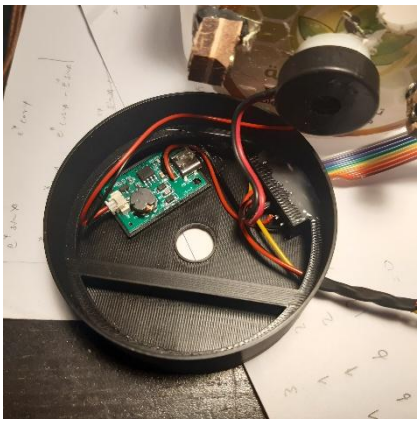
Water Handling Unit:

The Water Handling Unit (WHU) is a component of the Modular Greenhouse that resides within the plant enclosure and is connected to the main module via 7 wires of a ribbon cable. The WHU handles all water related systems, these are; water pump, ultrasonic atomizer and a capacitive soil moisture sensor.

A cube shaped plastic container with a screw on plastic lid (previously used to store honey) is used as the base of the system. Then a 3D printed hat for the lid is used to house the drive circuit of the atomizer, and to route the cables from the JST connector to the pump, atomizer driver PCB and the capacitive moisture sensor.

There are also three holes on the side of the hat, one for routing cables to the moisture sensor, the other two to allow for two 4mm screws to connect to anchors stuck on the lid using epoxy. The anchors are small wood pegs with a brass threaded insert for the screw to be driven into to hold the hat. At the top there’s a hole for the atomizer to spray the mist out of while there is a cotton (humidifier) swab that draws up water to the atomizer’s disk from the container.

A 12 V water pump resides within the container with its wires routed through the lid to the hat for power delivery. Anchoring of the pump is done using a bracket provided with the pump which is secured to the container with 4mm nuts and bolts. The pump used is submersible, so to deliver water out of the container there is a tube which goes through the top of the container (not the lid) to then be guided to the pot of the plant.





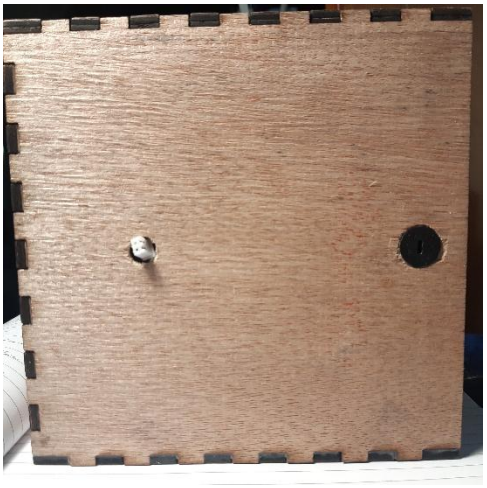
The Air Handling Unit

The Air Handling Unit (AHU) is another component of the Modular Greenhouse that sits beside the main unit and connects via 2 pins for the fan and a thin metal rod to the servo within the main module. The unit has grates that move along 3D printed rails using pegs attached to the grates. A rod from the main unit is guided through a slot on the AHU to connect to the first grate which is linked to all the other grate using stiff steel wire to move all the grates simultaneously.



AHU with grates in closed position

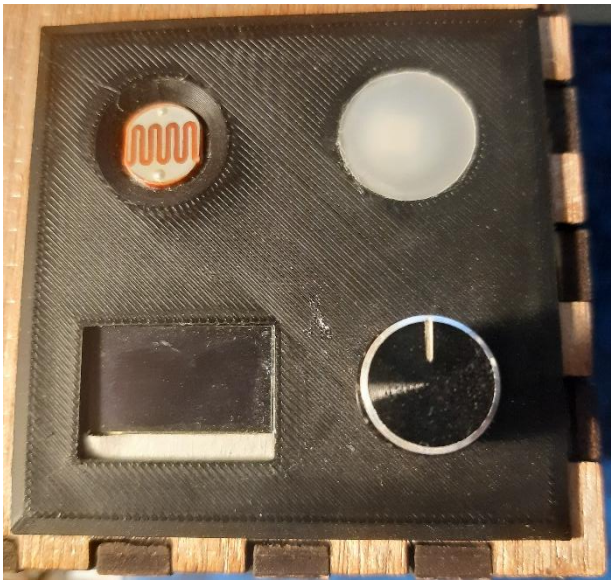
AHU with grates in open position



The left side has a hole to pass the JST connector for the fan. On the right side is the slot to take the steel wire that is driven by the servo to then open and close the grates.

The User Interface:

The User Interface (UI) was designed around primarily and OLED screen and a rotary encoder with a RGB LED for status to give users easy information on which system is currently active (Heating, cooling, etc.). On the panel holding these components is also a light dependent resistor (LDR) which was placed there for getting a read on light levels in the room to then control the grow light.



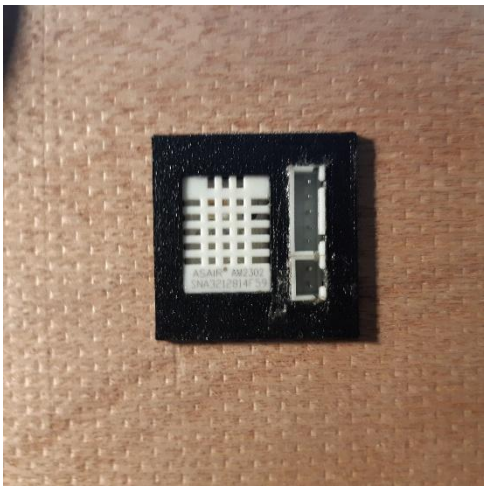
Top right sits the status LED which is covered by white plastic to diffuse the light.

On the top left is the LDR.

At the bottom right is the rotary encoder with which the user can navigate the menu and set parameters.

Bottom left sits the OLED screen, the white tape at the bottom is there to cover some accidental damage caused to the screen when setting it into the 3D printed panel.

The Peripheral Ports:

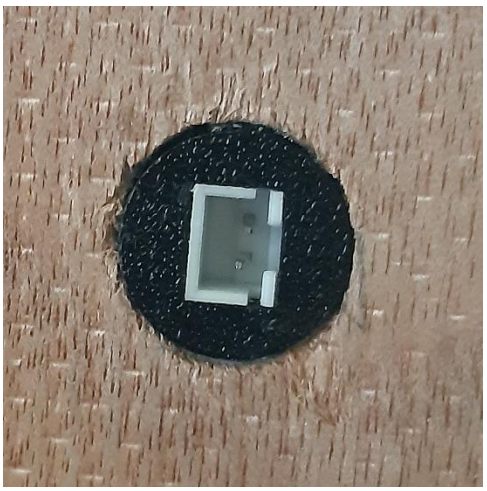


There are 4 distinct peripherals to the main control unit, those being; the WHU (Water Handling Unit), grow light, heat mat and AHU (Air Handling Unit).

The WHU connects through 7 JST pins which are shown here next to DHT22. Due to not having access to a singular 7 pin socket, I was forced to compromise and use a 5-pin with a 2-pin connector. Here the ribbon cable with JST connector attaches to connect to the WHU.



Here are the connectors to the heat mat and grow light. Both ports have a cover with a magnet press fit inside to easily open them. I am aware that XT60 connectors are meant for DC, but for this purpose I assumed that they would still work.



Here is the power connector that leads to the AHU. Just a 2-pin JST connector for the ground and 12 V DC used by the fan. One just needs to connect this with the cable at the side of the AHU followed by the servo rod for the grates and the system will be ready for operation.

Firmware

Custom Classes:

The Unit runs on micro-Python to manage the systems. I decided to make unique classes for most systems to make programming more streamlined. This has meant that there are 5 custom classes to handle the components and another class to make objects for the parameter's temperature and humidity.

These classes practically serve as wrappers for preexisting classes and to add special convenient functions to be used in main. For instance, "ledClass" builds on "neopixels" library but also adds a function to fade LED with specific color.

Main Loop:

The start of the script defines variables and sets up objects from classes.

At the start, current time is registered to be used by other methods in the cycle, ensuring it stays consistent through the loop.

Next are methods to detect a rotation of the encoder, followed by OLED display sleep logic where if there has been inactivity for over 10 seconds the OLED display switches off.

Servo update logic is next, where if there is a servo flag, the servo remains on for 0.4 seconds to get into position before turning off to save energy.

Next is logic to control output devices such as LED, pump, grow light, atomizer and servo. For the grow light, I set up logic that after an hour, if the grow light is on, it is shut off to then get a reading on the ambient light without the influence of the grow light. Based on the reading from LDR, the light will turn back on or stay off if ambient light exceeds user set threshold. There are also buffer regions added to other systems such as the fan, which turns on when temperature exceed threshold by more than two degrees and won't turn off until temperature drops to 1 above threshold.

Sensor readings come after this, the commands sit within a "try" so that if there is an error, it wont lead to the whole system crash.

Next is the UI script where I gave each menu a number and using if statements to determine which menu to display. Menu numbering is based on parent number and child number. The child's menu number is (parent menu number) + (child menu number among siblings), note there is concatenation, not actual addition. For instance, a parent menu numbered 12 has 3 submenus, so the submenus numbers become 121, 122, 123.

Due to knowledge ahead of time on the number of menus and submenus I made each decimal signify a menu. Though this system will likely get confusing if a menu has access to more than 10 submenus (10 unique integers between 0 and 9 (including 0 and 9)).

The rest of the code is just defining each menu and what it displays and/or controls (such as testing peripherals) Each menu has a timeout which if exceeded will switch the active menu back to its parent after 5 or 10 based on the menu.

Bill of Materials

Product	Price	Price (adjusted/used)
C13 male connectors	$4.02	$4.02
C13 female connectors	$4.54	$4.54
Power Supply 5V	$3.41	$3.41
Power Supply 12V	$4.40	$4.40
Heat Mat	$8.77	$8.77
Grow Light	$10.21	$10.21
Ultrasonic Atomizer + Driver circuit	$2.85	$2.85
KY-040 Rotary encoder (2 pcs)	$2.44	$1.22
Water Pump 12V	$4.44	$4.44
Heat Shrink set (127)	$1.26	~$0.10
Electronic components Kit (600pcs)	$9.98	~$0.20
JST XH set (230pcs)	$2.18	~$0.75
120mm Fan	$3.87	$3.87
RP2040 Zeros	$2.50	$2.50
WS2812 RGB LED (10 pcs)	$4.09	$0.41
Capacitive Soil Moisture Sensor	$1.95	$1.95
4mm Wood Threaded Inserts (20 pcs)	$2.00	$0.20
1 meter (8 Wire) Ribbon	$2.75	$2.75
0.96” OLED Display (white)	$2.40	$2.40
DHT22	$2.52	$2.52
Plywood total approximate	~$5.00	~$5.00
3D printing Filament (~100 g) Black PLA	~$4.50	~$4.50
XT60 Connector pair (5 pcs)	$6.65	$2.66
Wire + Solder and other miscellaneous	~$2.00	~$2.00
Perfboard 50 * 70mm	~$0.35	~$0.35
Totals	$99.17	$76.02

A Look Back:

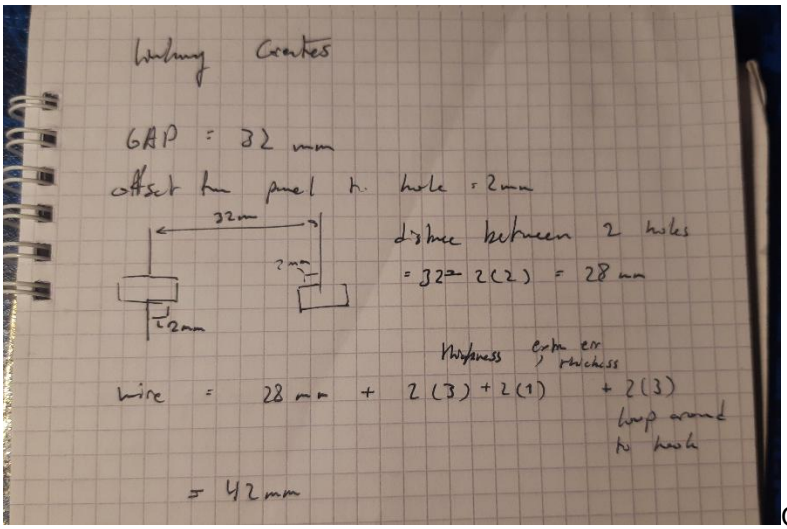
Now this project has many flaws, but it was first conceived over a year ago when my knowledge of the subjects involved was a lot less than it is today. If I were to make a new version I would;

- Ditch laser cut plywood, it warps and is difficult to work with at this scale, in the future I may instead use 3D printed container with built in holes to have threaded inserts.
- Better code: though micro-Python is easy and straightforward, in the future I would use C# instead and an MCU with Wifi capabilities such as esp32.
- More appropriate components such as proper connectors for the connecting the Modules and powering the peripherals running on mains voltage.
- For code, a more well thought out structure for easier expansion and perhaps even using different cores for different tasks for better speed.

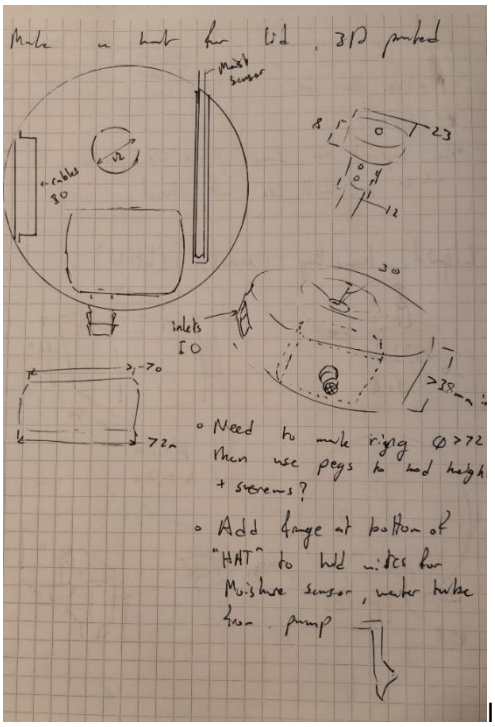
The time has come to close this project, if I were to try and make it perfect, I would never finish it. And honestly, I wish to get to work on other things and I'm satisfied with this, it is a prototype after all.

Extra pictures on the process:

Here are some pictures of notes I made while making this project, I thank you for reading and I hope that you have benefitted from this project.



Calculating wire length to connect the grates.



Initial Water Handling Unit sketches, this design did not work and was replaced by another design, which is the one found among the stl files.

Non blocking menu system

- Designate each menu w/ number (ID)

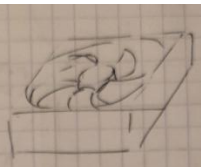
Then based on variable "active menu" system will show data relevant to that ID

Structure
- Main
  - (1) temperature
  - (2) humidity
  - ⋮
  - (9) Test
    - (91) Fan
    - (92) LED
    - (93) Servo
    - ⋮
    - (97) Pump
    - (98) exit

then just use if statement to show relevant info and to exit menu after set time of inactivity

also make it so screen update only after from encoder to make system snappier

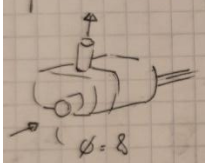
Fan orientation

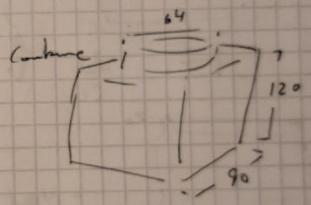


Notice that are no supports

The support on same plane as panel to which it will be attached

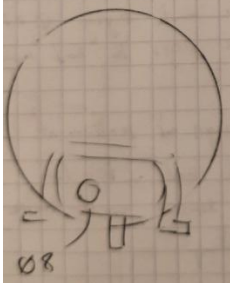
liquid handling unit (LHU)

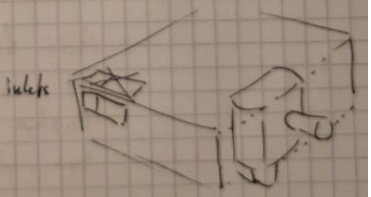


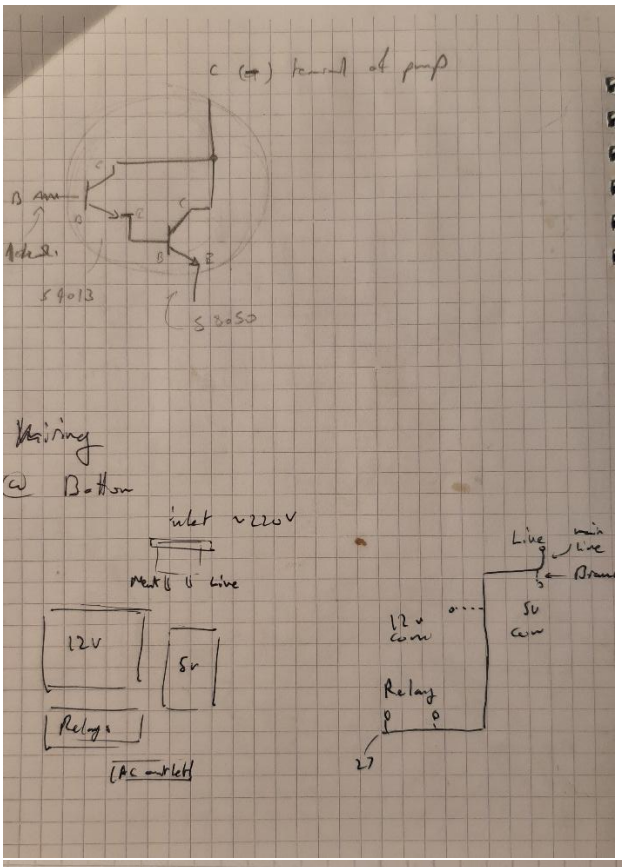


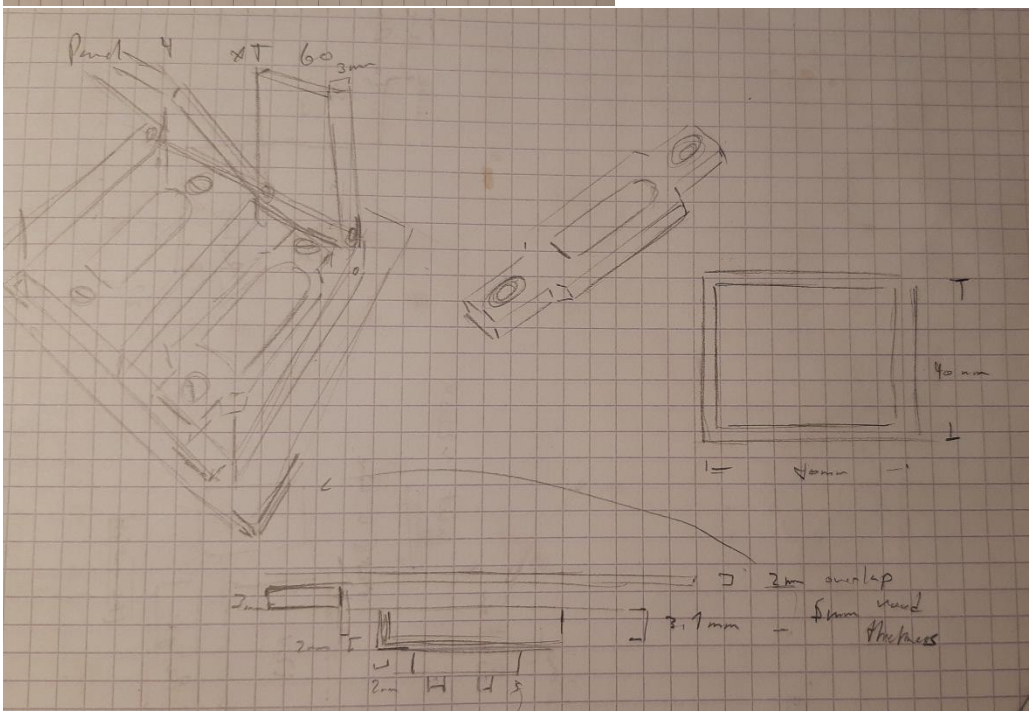
Lid Ø = 70

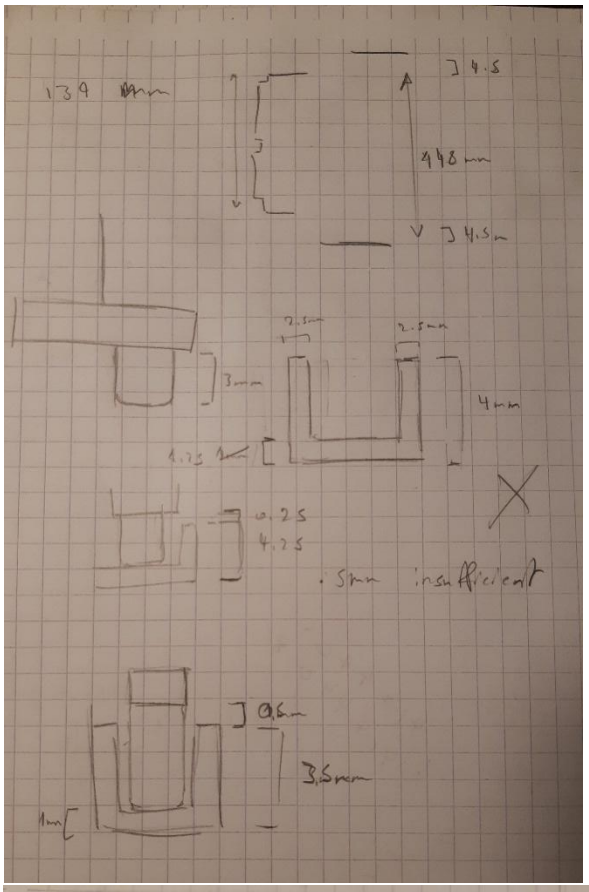
Lid h = 20

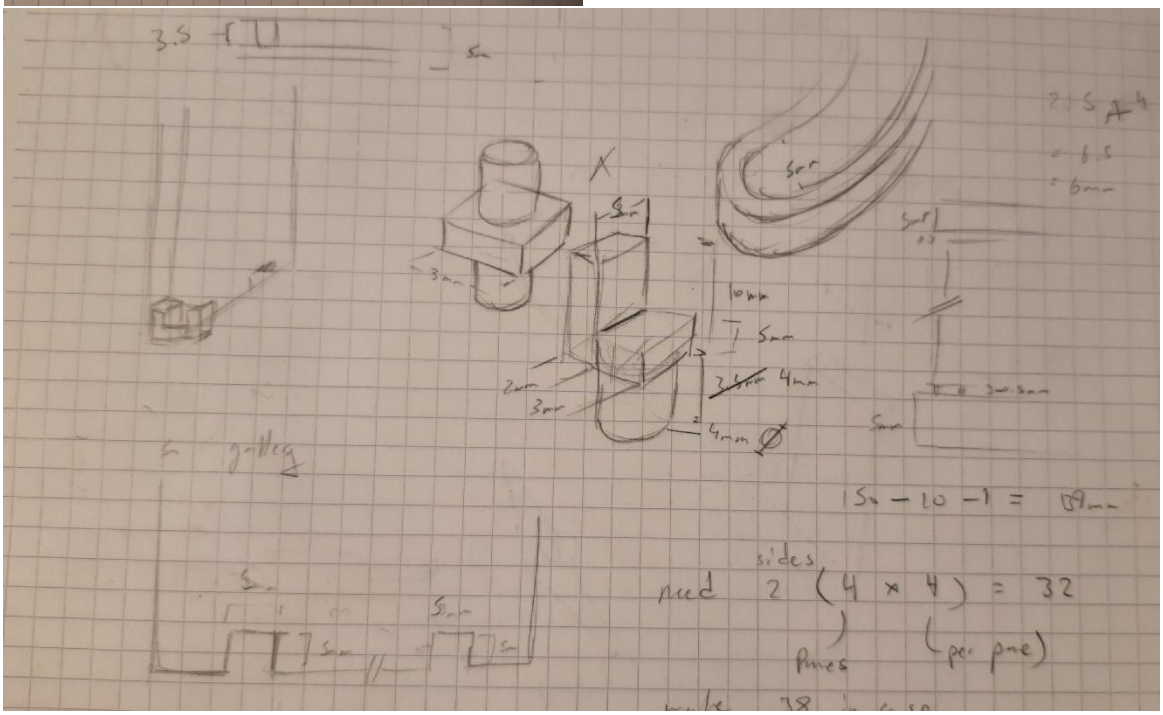


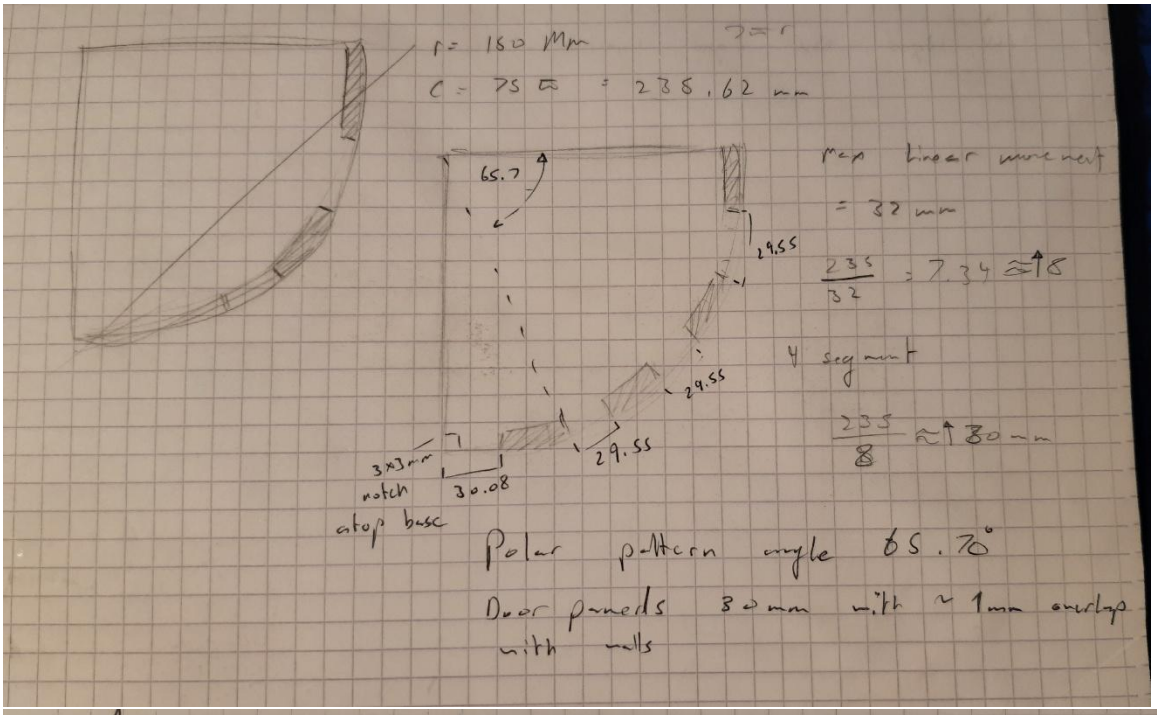


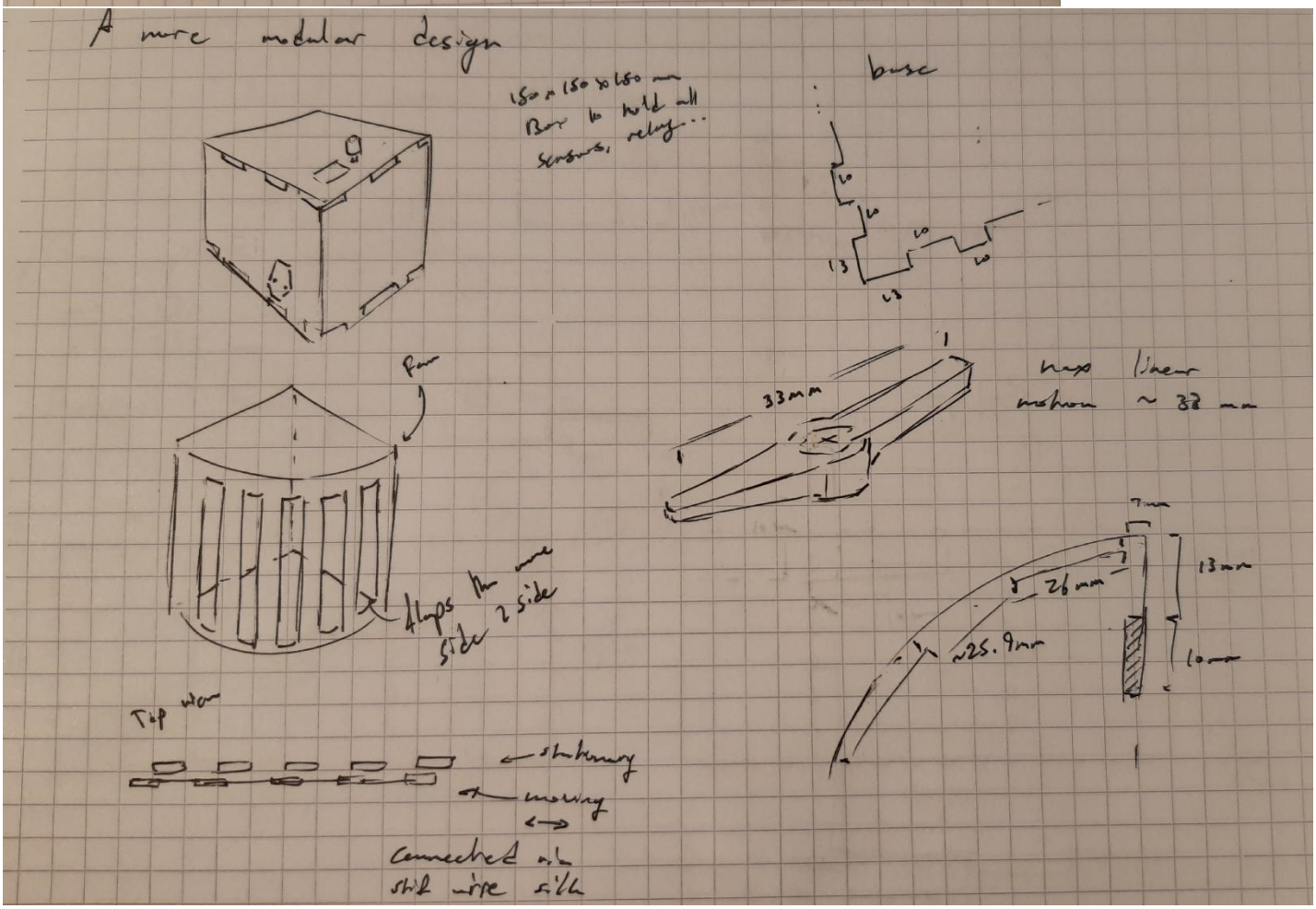

C (⇒) terminal of pump
B 4mm
Note:
S9013
S8050
Wiring
@ B-Hom
Inlet ~220V
Live
12V
5V
Relay
AC outlet
Live
main Live
Hom
12V com
5V com
Relay
27


Punch 4 XT 60 3mm
T
40 mm
2mm overlap
3.1mm
thickness
2mm
2mm




3.5
5mm
5mm
3mm
10mm
5mm
2mm
3mm
4mm
4mm
5mm
5mm
5


r = 150 mm
C = 75 π = 235.62 mm
65.7
29.55
29.55
29.55
3x3mm notch atop base
30.08
max linear movement = 32 mm
235/32 = 7.34 ≈ 8
4 segment
235/8 ≈ 30 mm
Polar pattern angle 65.78°
Door panels 30 mm with ~1mm overlap with walls


A more modular design
150 x 150 x 150 mm Box to hold all sensors, relays...
base
13
33mm
max linear motion ~ 32 mm
7mm
13mm
26mm
~25.9mm
10mm
Top view
4 loops for wire
side 2 side
stationary
moving
connected w/ slit wire sill